

Polymer Composites

- Isotropic – most metals, plastics and ceramic components are broadly isotropic.
- Anisotropy – with composites, processing route and final component properties are intertwined
- Properties depend on material orientation, hence the selected processing route.

Structure

- 3 constituent parts: Fibres, Matrix, Interozace

→ Fibres include: glass, aramid (Kevlar), Carbon

→ Matrix include: thermoset + thermoplastic resins – PE, epoxy, PEEK, Polyamide

→ Interozace altered by surface condition of fibres and processing conditions used for manufacture

Thermosetting Matrix:

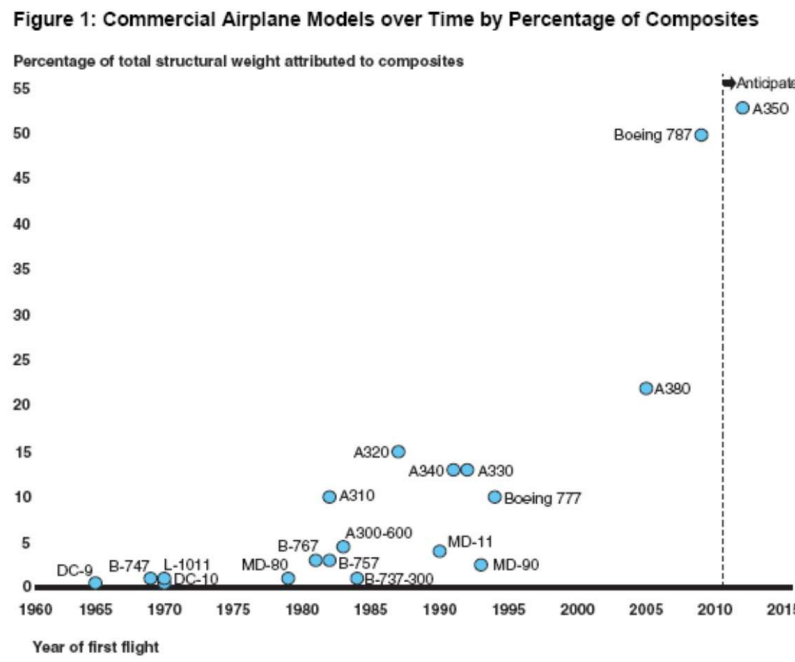
- Resin for matrix = Low viscosity liquid
- Resin flows and combines with solid fibres relatively easily.
- Fibres available in different forms
- Then Matrix undergoes curing reaction.

Common resin types:

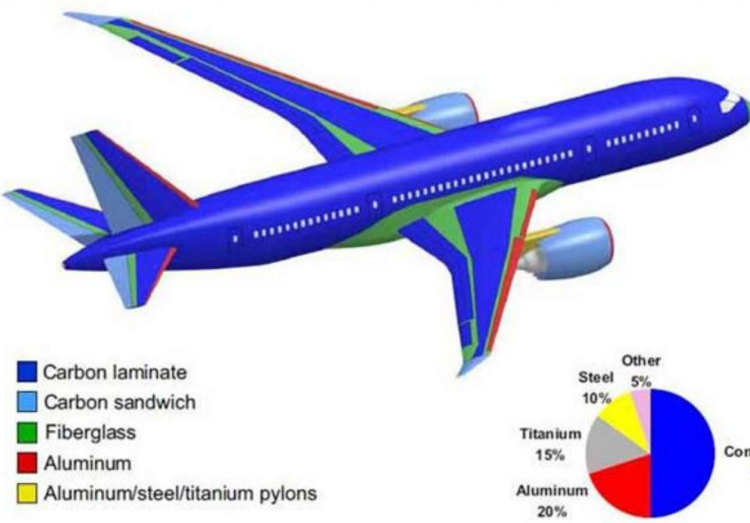
- Unsaturated polyesters
- Vinyl esters
- Epoxies
- Bismaleimides
- Polyimides phenolics
- Furanes
- Urethane - acrylates

Thermoplastic Matrix:

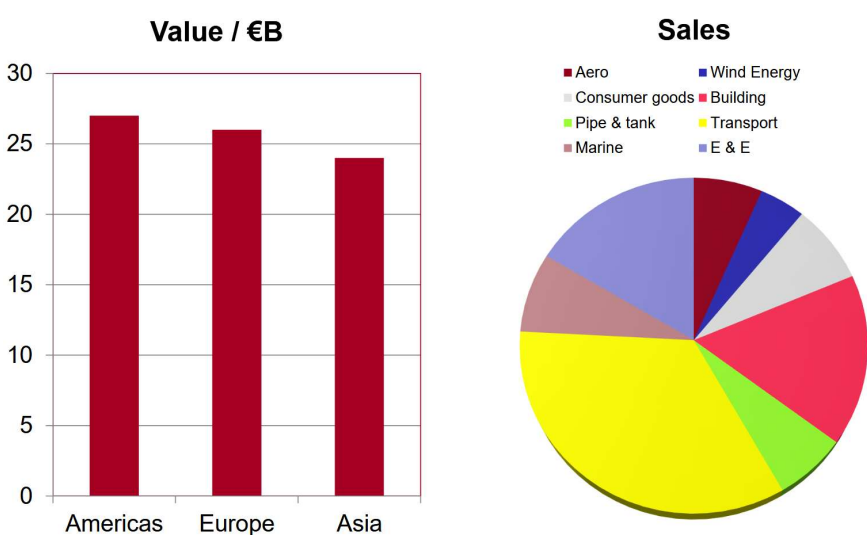
- Require heat to soften polymer matrix during processing
- The melt viscosity is typically higher than the resin viscosity of thermosets
- ∴ Difficulty in impregnating fibres with polymer; historically used in conjunction with short chopped fibres
- Resulting properties of composites are limited



Modern Aircraft Boeing 787 - Dreamliner



The Composites Industry



THE USA Composites market

- American composites market is 36% of the global composites industry in value (22 billion Euros) and 35% in volume (2.7 million metric tons).
- Composites in the USA represents 16% of the US structural materials market (compared with steel: 76%, aluminium: 8%)
- The volume of composites per capita is 8-9kg, compared to 1-2kg per capita in China, for instance.

Continuous roving

- Relatively cheap + consist of 1000's of fibres which can be aligned.

Chopped glass fibre mats

- Low cost random fibre composites

Continuous filament fibre mats:

- Similar to chopped; slightly better properties.

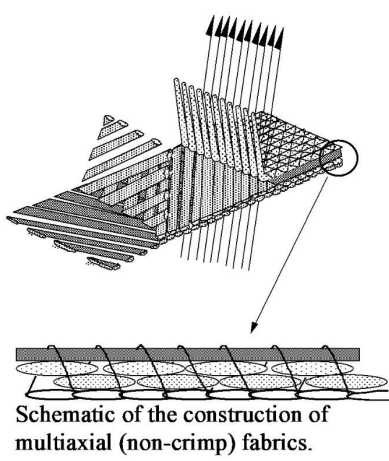
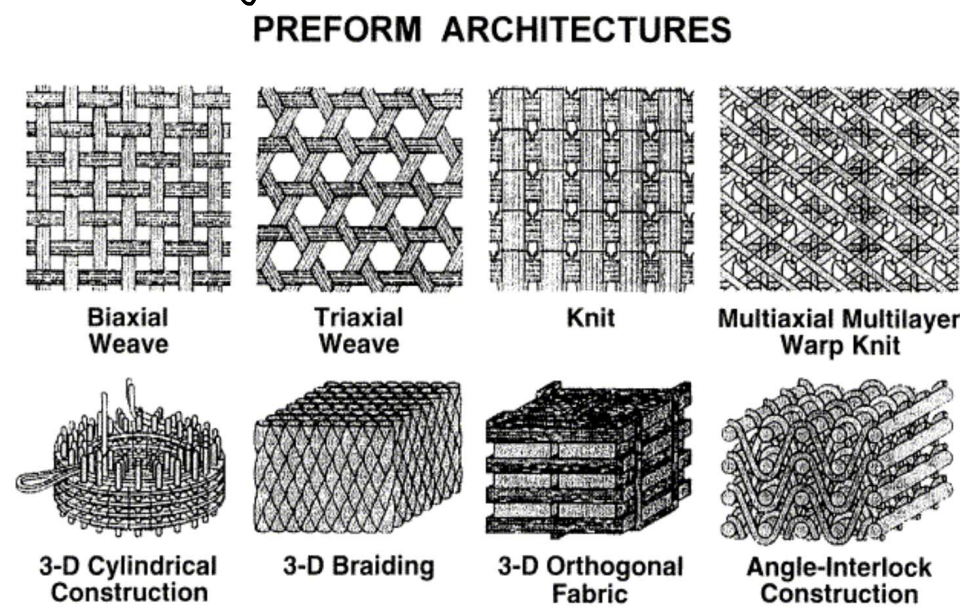
Woven fabrics:

- Higher Volume fraction, better properties

→ Triaxial Weaves

→ Adopting non-woven technology and stitching allows unidirectional or multidirectional continuous fibre fabrics to be produced without fibre crimping

→ Knitting + braiding techniques can produce 3D shaped preforms



Contact Moulding – Hand Lay Up Methods

1. Prepare mould- coat with silicone-free wax, to allow demoulding
2. Apply Gel-coat
3. Position fibres in (or over) mould
4. Wet-out fibres with resin (a low viscosity resin system is required)
5. Applied by brush
6. Roll to remove air and consolidate
7. Allow to cure
8. De-mould
9. Post-cure if necessary

Hand Lay Up

Advantages of Hand lay-up:

- Simple
- Low capital costs
- Large size

Disadvantages of Hand Lay up:

- High labour content
- Low speed
- Quality dependant on operator skill when laying out fibres
- Only one 'good' surface
- Fibres may be damaged
- Difficult to accurately position fibres

Typical products types include:

- storage tanks,
- chemical plant,
- boat hulls, and
- train cabs

Roll Add resin Reinforcement Mould Hand lay-up

Contact Moulding – Spray Up Methods

- Like hand lay up except the fibres and resin are deposited onto the mould by a spray system.
- Fibres are used in the form of continuous rovings, and the resin is mixed in the spray head.

Advantages of Spray up

- Portable
- Uses rovings (cheaper than mats)
- Capital investment low
- Fast
- Reduced labour costs
- Accurate metering of catalyst accelerator

Resin tank Fiber chopper Spray up Mould

Spray Up

Disadvantages of spray up

- Only one 'good' surface
- Only useful for relatively large areas
- Only random fibres, and
- Uniformity very dependant on operator.

Typical products types include:

- Baths
- Cladding panels
- Large boat hulls, and
- Fish ponds.

Baths Cladding panels Large boat hulls, and Fish ponds.

Resin Transfer Moulding (RTM)

- The dry reinforcement and any inserts are added to a mould made from light alloy or a glass reinforced plastic.
- The carefully mixed resins are combined and pumped under a low pressure (2MPa) into the mould. It must have a low viscosity.
- The press provides a clamping pressure during mould filling.
- Usually done at room temperature but cycle times can be reduced by heating the tools.
- Ensuring that fibres retain their position in a tool and to allow rapid positioning of fibres has led to the increasing use of preformed fibre mats.

Resin tank Fiber mat Moulding

Advantages / Disadvantages of RTM

Disadvantages of RTM

- Worse strength and stiffness properties to those achieved using hand lay-up
- This results from low volume fractions of fibres in the final component and the poor wet out of the fibres.

Advantages of RTM

- Potential for high volume production
- Applicable to a wide range of sizes with low cost tooling
- Ability to combine different reinforcement forms easily with a degree of control on fibre orientation
- Ability to incorporate inserts
- Two good surfaces are produced

Filament Winding

- The filament winding involves passing bare fibres through a resin bath and winding or wrapping them around a mandrel.
- Fibre orientation is controlled by the relative speed of rotation of the mandrel and the speed of traverse along the mandrel by the feed eye.
- The resin cures to form the composite and the mandrel is removed.
- The fibres must be fully impregnated with resin.
- The high fibre volume fraction makes high strength parts.

Resin tank Fiber spool Winding

Filament Winding

The process limitations are mainly:

- a restriction on production rate,
- the formation of only one high quality surface, and
- the inability to wind into concave surfaces.

The need to remove mandrels after winding certain shapes such as spheres is largely overcome by the use of inflatable, sacrificial or washout tooling.

Uses: tubes, fishing rods, wind turbine blades, pole vaults.

Prepreg Autoclave Processing

- Most high performance composites as consumed by the aerospace or F1 are used in the form of prepregs where the fibres are pre-wetted by a partially cured resin system.
- Typically, these prepregs are continuous fibre unidirectional plies although fabric prepregs are also available.
- The manufacturing processes comprise a variety of methods of applying prepregs to a tool, followed by an initial consolidation and removal of air, with a final consolidation and cure.
- High performance fibres are used, such as carbon, S-glass and aramid at volume fraction levels, typically > 60%.

Prepreg Autoclave Processing

The prepregs are arranged on the tool by either a hand lay-up or a tape lay-up machine (similar in concept to filament winding).

The prepregs typically retain their position on a tool during the lay-up process by virtue of their inherent "lack".

The assembly is then completely consolidated and cured by placing it in an autoclave.

Prepreg autoclave processing setup.

Pultrusion

- Pultrusion is analogous to extrusion as used for metals and plastics.
- Continuous profile sections are produced by pulling fibres impregnated with resin through a heated steel die.
- This process involves an impregnation stage and a consolidation / forming operation.
- The rate of pultrusion is determined by the rate of curing of the resin within the heated tool.
- The section is then wound onto a drum or cut into required lengths.
- USES: beams, channels, ladders, bridges, corrosion resistant walkways.

Resin tank Fiber spool Pultrusion

Hot Press Moulding

- High volume production (>10,000 articles from one mould) can require the use of hot press (compression or injection) moulding in combination with specially formulated moulding compounds.
- For mass production, the cycle time must be short at around a few minutes per article.
- The cure and the forming technique require high temperatures, which accelerates cure and reduces resin viscosity and the forming pressure.
- These materials must contain all of the ingredients for the final composite.
- The resin must be pre-accelerated and catalysed. The compound must also possess a reasonable storage life.
- Fillers are also added to the compound to aid processing:
 - by making the material dry
 - reducing the shrinkage of the material after moulding and
 - to reduce the cost

Moulding

Compression moulding (typically uses sheet moulding compound)

- Flash is difficult to remove (contains fibres).
- Charge difficult to meter and position reproducibly.

Injection Moulding (typically uses dough moulding compound):

- The mould is closed and compound introduced via runners.
- Moulds/tools clamping devices very expensive.
- Part size limited by machine considerations.
- Fibres suffer attrition during transport along screw of injection moulder and during passage through runners and die.
- Probability of alignment of fibres due to flow geometry may result in unfavourably orientated weak sections.

High Performance Composite Structures

- High performance composites require continuous fibres positioned accurately in certain specified directions.
- The processes available depend on the initial form of the materials.
- For large and complex shapes the fibres and matrix usually come in a pre-combined sheet form known as prepreg which are processed in an autoclave or equivalent device.

High performance composite structure.

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